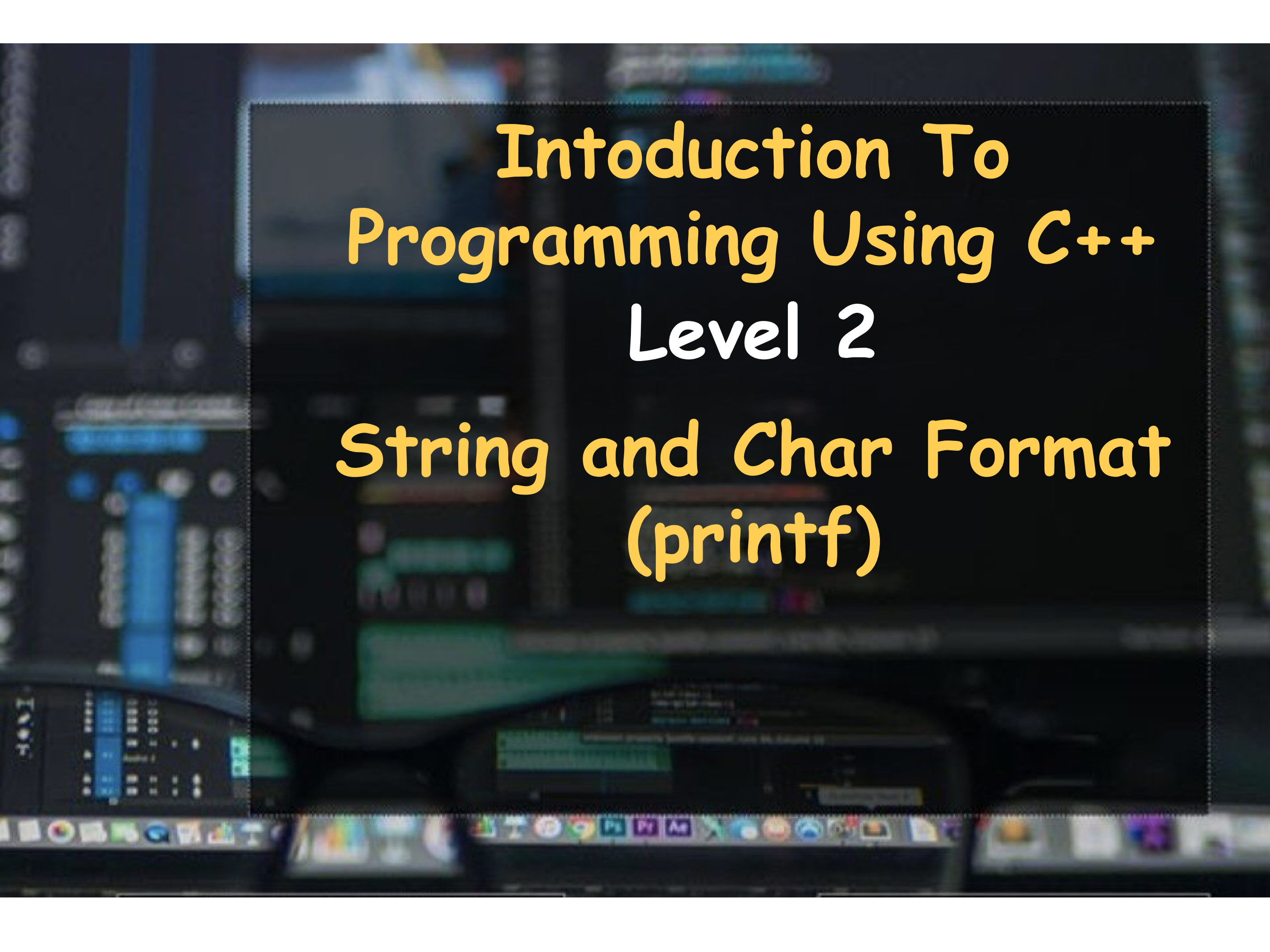
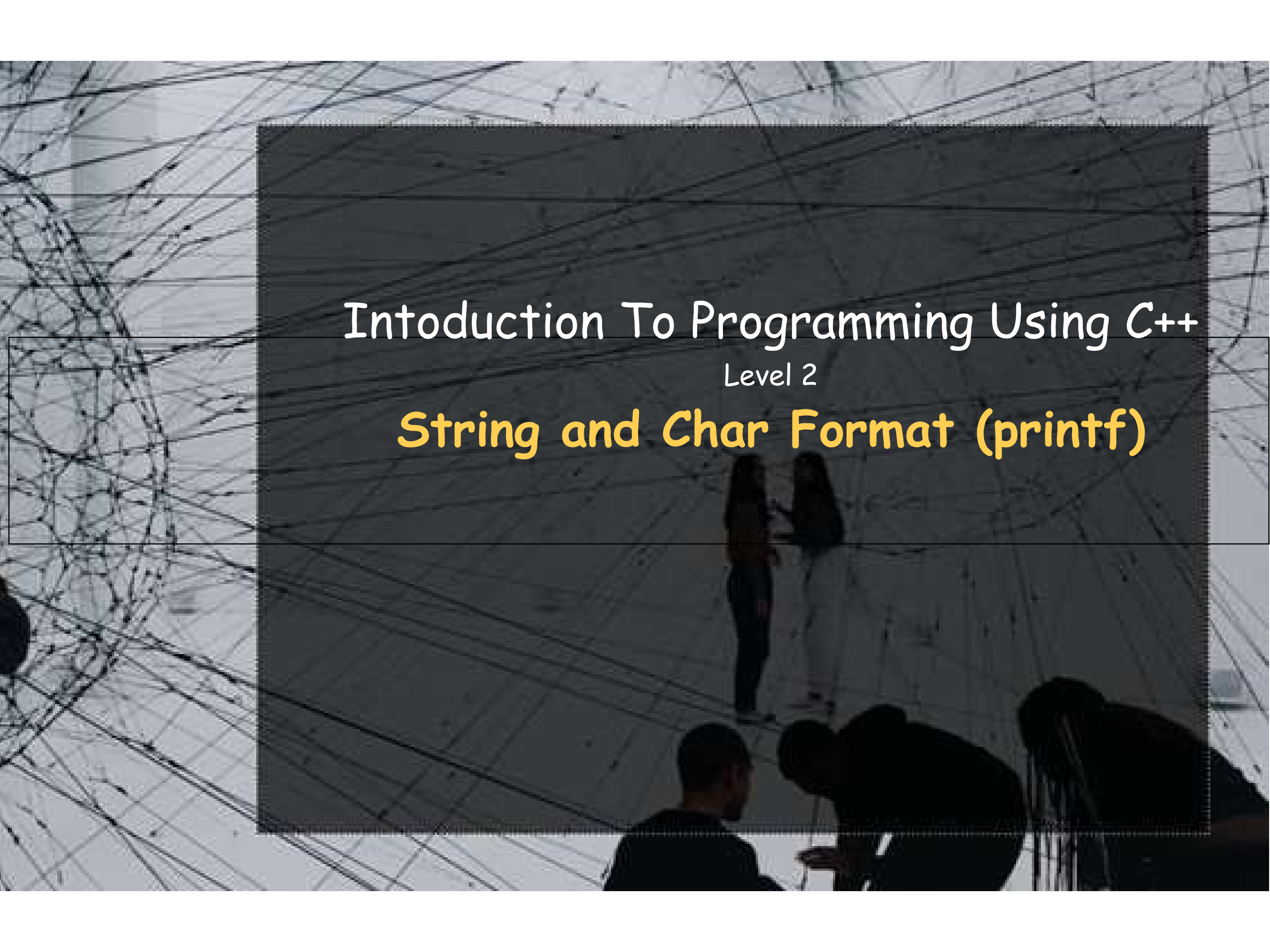


The background is a blurred image of a computer monitor. It shows a dark-themed code editor with some text in green and blue. At the bottom, a Windows taskbar is visible with various application icons like Chrome, Photoshop, and others.

Introduction To Programming Using C++ Level 2 String and Char Format (printf)



Introduction To Programming Using C++

Level 2

String and Char Format (printf)

Definition

- ◆ `printf()` is a formatted output function from the C Standard Library that can also be used in C++.
- ◆ When using `printf()` for text, `%s` is used to print a C-Style String, which is a character array ending with the null character `'\0'`.
- ◆ `%c` is used to print a single Character.

Explanation

- ◆ `printf()` does not directly understand the C++ `std::string` Type through `%s`.
- ◆ `%s` expects a pointer to a Null-Terminated Character Array, such as a `char[]` or a C-Style String.
- ◆ The `std::string` Type belongs to C++, so it should normally be printed using `cout`.

String Format - `%s`

1. `%s`

▣ `%s` is used to print a C-Style String until the Null Character `'\0'` is reached.

```
char Name[] = "Sara Alterkawi";  
printf("Dear %s, How are you?\n", Name);
```

➔ Result

Dear Sara Alterkawi, How are you?

▣ Format Structure:

`%s`
└── String
└── Format begins

2. `%.3s`

▣ `%.3s` limits the maximum number of Characters printed to 3.

```
printf("%.3s\n", Name);
```

➔ Result

Sar

▣ Here, `3` specifies the maximum number of Characters to print.

3. `%.\*s`

▣ `%.\*s` allows the maximum number of Characters to be specified dynamically through an argument.

```
printf("%.*s\n", 3, Name);
```

➔ Result

Sar

Character Format - `%c`

1. `%c`

▣ `%c` is used to print a single Character.

```
char c = 'S';  
printf("%c\n", c);
```

➡ Result
S

2. `%\*c`

▣ `%\*c` allows the Field Width to be specified dynamically through an argument.

```
char c = 'S';  
printf("Setting the width of c :%*c\n", 3, c);
```

➡ Result

Setting the width of c : S

▣ The Width is `3`, and because `S` occupies one position, two spaces appear before it.

💡 Example

```
int main()
{
    char Name[] = "Sara Alterkawi";
    // Print C-Style Strings
    printf("Dear %s, How are you?\n\n", Name);
    // Print a Character
    char c = 'S';
    printf("Setting the width of c :%*c\n", 1, c);
    printf("Setting the width of c :%*c\n", 2, c);
    printf("Setting the width of c :%*c\n", 3, c);
    // Limit the number of Characters
    printf("First 3 characters: %.3s\n", Name);
    return 0;
}
```

➡ Result

```
Dear Sara Alterkawi, How are you?
Setting the width of c :S
Setting the width of c : S
Setting the width of c :  S
First 3 characters: Sar
```

Note

- ◆ A C-Style String is a Character Array terminated by the Null Character `'\0'`.
- ◆ Internally, this contains:
S a r a \0
- ◆ `printf("%s", Name)` keeps printing Characters until it reaches `'\0'`.

⊘ Warning

- ◆ Do not pass a C++ `std::string` directly to `printf("%s", ...)`.
 `string Name = "Sara";`
 `printf("%s", Name); // Incorrect`
- ◆ `%s` expects a C-Style String, not a `std::string` object.
- ◆ If you have a `std::string`, you can use `.c_str()`:
 `string Name = "Sara";`
 `printf("%s", Name.c_str());`
- ◆ Or, in normal C++ code, simply use:
 `cout << Name;`

Comparison

▨ `%.3s`

- ✓ The maximum number of Characters is fixed.
- ✓ Example: `printf("%.3s", Name);`
- ✓ Always prints up to 3 Characters.

▤ `%.\*s`

- ✓ The maximum number of Characters is dynamic.
- ✓ Example: `printf("%.\*s", 3, Name);`
- ✓ The number is provided as an additional argument.

▨ `%c`

- ✓ Prints one Character.
- ✓ Example: `printf("%c", c);`

▤ `%\*c`

- ✓ Prints one Character with a dynamically specified Field Width.
- ✓ Example: `printf("%\*c", 3, c);`

✦ Summary

- ✓ ``%s`` is used to print C-Style Strings.
- ✓ A C-Style String is a Character Array terminated by ``'\0'``.
- ✓ ``%.3s`` limits the output to a maximum of 3 Characters.
- ✓ ``%.*s`` allows the maximum number of Characters to be specified dynamically.
- ✓ ``%c`` is used to print a single Character.
- ✓ ``%*c`` allows the Character's Field Width to be specified dynamically.
- ✓ ``printf()`` does not directly accept ``std::string`` with ``%s``; use ``c_str()`` or ``cout``.

Thank you 😊